

DOKUMENTASI TEKNIS

Architecture & Extension Guide

WEBSITE DESA MAKMURJAYA

Panduan teknis untuk pengembang lanjutan

Disusun oleh

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SECTION 01

Architecture Overview

How the pieces fit together

This is a static site built with Astro 4.x, styled with Tailwind CSS, content-managed via Decap CMS (Git-backed), and hosted on GitHub Pages (free). All content lives as Markdown/JSON files in the same Git repository as the source code.

Request & Publish Flow



💡 Ringkas

Admin edits in CMS → Git commit → GitHub Actions builds → static files deployed to the GitHub Pages CDN. No server, no database — everything is a file in Git.

SECTION 02

Technology Stack & Justifications

Why each tool was chosen

Technology	Role	Chosen over
Astro 4.x	Static site generator. Zero JS by default; Islands Architecture for interactive components.	Next.js (overkill for static), Hugo (less flexible templating)
Tailwind CSS 3.x	Utility-first CSS with custom design tokens in tailwind.config.mjs.	Bootstrap (too opinionated), plain CSS (maintenance burden)
Decap CMS 3.x	Browser-based, Git-backed. No server needed.	WordPress (needs hosting), Strapi (needs backend), Sanity (paid at scale)
GitHub Pages	Free static hosting with CDN.	Netlify / Vercel (simpler, but adds account sprawl)
Leaflet.js	Open-source maps. No API key needed.	Google Maps (requires billing)
Pagefind	Client-side search built at deploy time. Zero runtime cost.	Algolia (paid)
Vitest + fast-check	Unit + property-based testing. Fast, ESM-native.	—
TypeScript	Type safety for maintainability. Strict mode enabled.	—

SECTION 03

Project Structure

A map of the repository

src/	
├ components/	→ Astro components (by feature)
│ └ common/	→ Header, Footer, Navigation, SEOHead
│ └ beranda/	→ Homepage sections
│ └ berita/	→ BeritaCard
│ └ pengumuman/	→ PengumumanCard
│ └ umkm/	→ UMKMCard, UMKMFilter
│ └ wisata/	→ WisataCard, WisataDetail, WisataGaleri
│ └ pemerintahan/	→ OrgChart, PerangkatCard
│ └ galeri/	→ GaleriGrid, Lightbox
│ └ kontak/	→ KontakForm, PetaKantor
│ └ profil/	→ ProfilNav
│ └ ui/	→ Pagination, Badge, EmptyState, Accordion
├ content/	→ Astro Content Collections (Markdown/JSON)
│ └ berita/	→ News articles (.md)
│ └ pengumuman/	→ Announcements (.md)
│ └ wisata/	→ Tourism destinations (.md)
│ └ umkm/	→ UMKM businesses (.md)
│ └ galeri/	→ Photo gallery (.json)
│ └ perangkat-desa/	→ Government officials (.md)
│ └ faq/	→ FAQ entries (.md)
│ └ halaman-statis/	→ Static pages (.md)
│ └ infografis/	→ Data JSON (penduduk, idm, apbdes, etc.)
├ data/	
│ └ situs.json	→ Site name, contact, hero text (CMS-editable)
├ layouts/	→ BaseLayout, ContentLayout
├ pages/	→ File-based routing
├ styles/	→ global.css (Tailwind + component classes)
└ utils/	→ date, currency, pagination, etc.
public/	
├ admin/	→ Decap CMS (index.html with inline config)
├ uploads/	→ Media files uploaded via CMS
└ robots.txt	

SECTION 04

Data Flow

Where content lives and how it moves

Content Collections (Markdown)

Berita, Pengumuman, Wisata, UMKM, FAQ, Perangkat Desa, and Halaman Statis use Astro Content Collections defined in `src/content/config.ts`. Each collection has a Zod schema for type validation at build time.

Infografis Data (JSON)

Infografis data lives in `src/content/infografis/*.json`. These are imported directly in `pages/infografis.astro`. Decap CMS edits these files via the 'files' collection type.

Site Settings (JSON)

Global site text (nama desa, alamat, tagline) lives in `src/data/situs.json`. Imported by Header, Footer, and Homepage components. Editable via the CMS 'Pengaturan Situs' menu.

`getPath()` Utility

All internal links use `getPath('/path')`, which prepends the base URL (`/website-desa-makmurjaya/`) for GitHub Pages subdirectory deployment.

Penting

Never hardcode paths — always use `getPath()`. Hardcoded links will break the moment the base path changes.

SECTION 05

How to Add a New Page

Example: adding a /layanana page

1 Create src/pages/layanana.astro

2 Import BaseLayout and getPath:

```
---
import BaseLayout from "../layouts/BaseLayout.astro";
import { getPath } from "../utils/links";
---
<BaseLayout title="Layanan" description="...">
  <div class="page-header">...</div>
  <div class="max-w-7xl mx-auto px-4 md:px-8 py-8">...</div>
</BaseLayout>
```

3 Add a navigation link in src/components/common/Header.astro (both desktop navlinks and the mobile drawer).

4 Build and test: npx astro build

SECTION 06

How to Add a New CMS Collection

- 1 Define the schema in `src/content/config.ts` using Zod.
- 2 Create the folder: `src/content/your-collection/`
- 3 Add the collection to the Decap CMS config in `public/admin/index.html`
- 4 Create the page that queries this collection using `getCollection()`

Penting

The collection name in `config.ts` must match the folder name and the Decap CMS collection name exactly.

SECTION 07

How to Add a New Infografis Tab

- 1 Create `src/content/infografis/your-tab.json` with initial data.
- 2 Add a CMS file entry in `public/admin/index.html` under the 'infografis' collection.
- 3 In `src/pages/infografis.astro`: import the JSON, add a tab button, add a panel div.
- 4 The tab-switching JS is already generic — it works on `aria-controls/role` attributes.

SECTION 08

How to Change the Design System

Everything flows from tailwind.config.mjs

All design tokens are defined in tailwind.config.mjs:

- **Colors** — primary (greens), gold (accents), ink, muted, paper, line
- **Fonts** — font-display (Lexend), font-body (Inter)
- **Shadows** — card, elevated

Component Classes

Defined in src/styles/global.css under @layer components:

- .card, .stat-card, .pill, .btn-primary, .btn-outline, .btn-ghost
- .page-header, .tab-bar, .tab-item, .empty-state, .page-btn
- .progress-wrap, .progress-fill, .table-clean

Tips

To change colors, update tailwind.config.mjs — every component updates automatically.

SECTION 09

Deployment & CI/CD

Fully automated via GitHub Actions

Pipeline defined in `.github/workflows/deploy.yml`:

- 1 Push to the main branch
- 2 GitHub Actions triggers: checkout → Node 20 → npm ci → astro build → pagefind → upload artifact
- 3 Deploy to GitHub Pages

💡 Ringkas

Build time: ~30–60 seconds. Live at <https://jeff146354.github.io/website-desa-makmurjaya/>

Environment Variables

Variable	Notes
FORMSPREE_ENDPOINT	Set in GitHub repo secrets (Settings → Secrets → Actions)
—	No other env vars needed for basic operation

Custom Domain Setup

- 1 Add the domain to public/CNAME
- 2 Update 'site' in `astro.config.mjs`
- 3 Update 'base' to '/' if using a root domain
- 4 Configure DNS (CNAME record pointing to `jeff146354.github.io`)

SECTION 10

Known Limitations & Tech Debt

- No real-time preview — the site rebuilds on each commit (1–2 min delay)
- Pagefind search index only updates on deploy, not on CMS save
- Image optimization happens at build time only (no runtime CDN resizing)
- No image upload size enforcement — the CMS only shows a hint (5MB)
- Formspree free tier: 50 submissions/month
- CMS requires a GitHub account — the village admin needs a GitHub login
- Decap CMS OAuth proxy depends on Netlify (external dependency)
- No offline editing capability
- No multi-language support (Indonesian only)
- Mobile bottom nav overlaps content at the bottom — 64px spacer added

SECTION 11

Recovery Playbook

When things go wrong

Build fails after a CMS edit

- 1 Check the GitHub Actions tab for the error message
- 2 Most likely cause: invalid frontmatter in a .md file (missing required field)
- 3 Fix: edit the problematic file via the CMS or directly in GitHub
- 4 Push the fix — the build auto-retries

Rollback a bad commit

```
git log --oneline    # find the good commit hash
git revert HEAD      # revert the last commit
git push origin main
```

Set up on a new GitHub account

- 1 Fork or transfer the repository
- 2 Enable GitHub Pages (Settings → Pages → Source: GitHub Actions)
- 3 Update the repo name in public/admin/index.html (backend.repo field)
- 4 Set up the Netlify OAuth proxy (or use the github-oauth-app alternative)
- 5 Update astro.config.mjs site & base URLs

CMS login not working

- 1 Verify the Netlify site_id in public/admin/index.html matches your Netlify app
- 2 Check the GitHub OAuth App callback URL
- 3 Ensure the GitHub user has write access to the repository